

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building AI agent systems in TypeScript/React/Node.js. At Isomer AI, built multi-model LLM action-call pipelines (MCP, Claude/GPT-4o/Gemini), real-time event architecture with Pusher, and 5+ third-party API integrations -- 394 commits across 7 production deployments. Architected AlgoPulse, an agentic platform with Temporal.io + Kafka fault-tolerant orchestration reaching 2,000 async inference req/sec. Full-stack builder with backend and distributed systems depth, ready to own end-to-end feature delivery from day one. Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Full-Stack TypeScript / React / Node.js

LLM Action Calls & Token Streaming

AI Agent Orchestration (MCP)

Distributed Systems (Kafka, Temporal)

Real-Time Architecture (WebSockets)

Third-Party API Integration

GraphQL APIs

Production Feature Delivery

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - TypeScript / Node.js / React

- Built **multi-model LLM action-call routing** with MCP across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models -- unified token-streaming interface with per-provider fault isolation, running in production
- Architected **real-time event pipeline** using Pusher + Redis pub/sub for live AI processing status updates; integrated Microsoft Graph API for email parsing, S3 for document storage, Auth0 for identity -- 5+ third-party integrations stable across 7 deployments
- Implemented **Temporal.io workflow orchestration** for multi-step claims automation: durable state machines, retry policies, and RBAC -- 394 commits to codebase, 10+ backend features shipped

NC State University Apr 2025 - Present

Graduate Research Assistant - Cloud Platform / ML

- Built cloud-based genomic data platform with JBrowse 2 on Google Cloud Run, enabling interactive exploration of 500GB strawberry and grape datasets for 100+ researchers worldwide
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML -- improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated Grafana monitoring dashboards; supported model validation and red-teaming

Capgemini Jun 2023 - Aug 2023

AI/ML Intern - Distributed Systems

- Implemented **distributed data pipelines** using Hadoop + Spark for batch processing, and Kafka streaming for real-time data ingestion -- reduced ticket-processing latency by 20%
- Led end-to-end SDLC for support ticket analysis system, reducing processing time by 15%

PROJECTS

AlgoPulse -- Agentic AI Platform Independent

Identified API-polling bottleneck under heavy inference load and re-architected to **event-driven streaming**: Supabase Edge Functions stream AI reasoning loops to client in real time. Temporal.io manages durable workflow state; Kafka handles backend message brokering -- **2,000 async inference req/sec without state loss**. Fault-tolerant worker architecture via Temporal state machines.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Privacy-First RAG System Independent

Built a compliance-ready RAG pipeline for fintech/healthcare with per-tenant data isolation at the vector retrieval layer -- prevents cross-tenant context leakage and LLM hallucinations. Deterministic output architecture for regulated data environments.

Python, LLMs, Vector Databases, RAG

PerCent -- GenAI Finance App HackNC24 Top 10 / 100+ teams

Full-stack GenAI finance app with goal tracking, expense logging, and Qwen-14B-Chat RAG chatbot -- built complete MVP in 48 hours, ranked Top 10. Reduced manual financial research time by 30%.

Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

Mapify -- ML & NLP Tool IEEE Published

T5-transformer pipeline converting 1,500+ PDFs/images into mind maps and flowcharts. Reduced academic review time by 60%; presented to 500+ attendees at IEEE conference.

T5-transformer, NLTK, NetworkX

EDUCATION

Master of Science, Computer Science North Carolina State University

Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms

B.Tech, Computer Science Mumbai University

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Warehousing, Big Data, DS & Algorithms, Machine Learning, Software Engineering, Blockchain

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C Frontend/Backend: React, Next.js, Node.js, Express, GraphQL, Prisma, Tailwind

AI/LLM: MCP, OpenAI API, Anthropic Claude, Gemini, LangChain, Hugging Face, RAG, PyTorch, TensorFlow

Distributed Systems: Temporal.io, Kafka, Spark, Hadoop, Redis, PostgreSQL, Supabase

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Terraform, Grafana, Sentry

Awards: National Hackathon Winner - Top 10 in 6+ hackathons; Google Developers Student Club Lead (500+ students)